Mingshuo Wang

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EDUCATION

University of Southern California
M.S. Petroleum Engineering & M.S. Engineering Management *May 2025*

Hunan University of Science and Technology
B.S. Chemical Engineering *July 2022*

WORK EXPERIENCE

KEM Power LLC, KEM Energy Inc. Houston, TX, US
Solar Project Engineer Intern *Jun – Sep 2024*

- Tracked *Sungrow* utility-scale solar project milestones by updating Gantt charts, reviewing weekly contractor reports, and flagging schedule risks to senior engineers.
- Prepared logistics plans for panel and inverter shipments (truck routing, warehouse coordination, and delivery sequencing) and presented rollout updates during client meetings.

Oil & Gas Engineer Intern *Jun – Sep 2023*

- Assessed gas measurement and control upgrades, contributing to proposals for upstream projects.
- Estimated costs and schedules for oilfield bids, improving proposal accuracy and contract competitiveness.

Champion Asia Group Ganzhou, China
Chemical Engineering Intern - Water Treatment *Jun – Aug 2021*

- Ran Aspen Plus simulations to optimize chemical dosing ratios in wastewater treatment, validating results with lab tests.
- Identified pump/pipe bottlenecks in the plant layout and proposed equipment upgrades that improved daily treatment capacity.

Hunan Lichen Industry Co., Ltd. Changsha, China
Chemical Engineering Intern - Quality Assurance *Oct – Dec 2020*

- Operated surfactant production systems including spray drying and thermal aging units, ensuring consistent output quality and adherence to process parameters.
- Conducted QA/QC testing and data analysis for production processes, ensuring compliance with safety and quality standards.

SKILLS

- *Software*: proficient in **Python, autoCAD, ArcGIS**, **COMSOL**, **MATLAB**, **R studio**, **SQL, Power BI** and **Aspen Plus**
- *Technical Skill*: Risk assessment, inventory control, logistics planning, safety inspections, KPI monitoring, and sustainability tracking. Experienced in supporting compliance and cross-functional operational initiatives. Data visualization, statistical analysis, probabilistic modeling, database management

PUBLICATION

- *Exploring Modern Feature Extraction Techniques for Improved Offshore Fault Detection in Oil and Gas Operations*
  R. A. Wibawa, **M. Wang**, B. Jha
  [Submitting to SPE ATCE 2025]

PROJECT EXPERIENCE

USC GEM-Lab, Funded by ARPA-E Dr. Birendra Jha
Research Assistant *Sep – Dec 2024*

- Involved in Advanced Research Projects Agency-Energy (ARPA-E) conducting multiscale characterization, transport, and mechanics for enhanced H2 recovery and reservoir management.
- Simulated coupled geomechanical-chemical processes using COMSOL to model serpentinization reactions, tracking permeability changes under varying stress/porosity—experience directly applicable to field-based geohazard analysis.

Deep Learning for Stock Price Prediction Using LSTM Dr. Zhenglu Li
Research Assistant *Mar - May 2024*

- Developed an end-to-end LSTM stock prediction pipeline using deep learning techniques, optimizing architecture with stacked recurrent layers and automated hyperparameter tuning, achieving 0.06 RMSE.

Simulation and Optimization of CO_2 Injection in Porous Media
Research Assistant *Jan – May 2023*

- Developed and implemented a MATLAB-based simulation framework for geologic CO_2 storage, incorporating uncertainty-aware risk assessment, history-matching for model calibration, and optimization of injection strategies.
- Evaluated and mitigated leakage with R and fracturing risks through data-driven adjustments to well placement and injection rates, ensuring efficient long-term CO_2 sequestration (leakage rate <0.1%).